

Adrian Diaz

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EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science

Atlanta, GA

Expected Graduation Date: May 2028

California State University, Fullerton

Bachelor of Science in Computer Science

Fullerton, CA

August 2024 - May 2026

EXPERIENCE

Graduate Research Assistant

June 2026 – Present

California State University, Fullerton

Fullerton, CA

- Engaged in an industry funded research project supported by American Honda Motor Co. focused on human-AI teaming and trust in autonomous systems for manufacturing environments.
- Developed Python simulation workflows in NVIDIA Isaac Sim and explored integration with ROS/ROS2 for robotics communication and coordination.
- Contributed to an ongoing research publication within the Design-is-ALL lab.

Software Engineer Intern

May 2026 – Present

Cubillite Cup

Remote

- Developed backend services for a Cuban dice game with 100K+ downloads using Firebase and TypeScript.
- Designed live gameplay architecture using Photon Fusion to synchronize player state across concurrent sessions.
- Utilized unit testing and CI/CD pipelines to validate gameplay logic and support stable releases across staging and production mobile builds.

Software Engineer Intern

July 2021 – August 2021

University of California, Irvine

Irvine, CA

- Developed Python code for autonomous motion planning, modeling navigation logic and pathfinding algorithms.
- Implemented Dijkstra's algorithm to improve route planning efficiency for obstacle rich environments.
- Visualized algorithm behavior using Matplotlib and mathematical libraries to identify performance bottlenecks and enhance motion logic.
- Contributed to detailed technical documentation on algorithm design, facilitating team knowledge sharing.

PROJECTS

Azure Cloud Migration | *Microsoft Azure, Windows Server, Active Directory*

February 2026 – May 2026

- Migrated a Windows Server 2022 Active Directory to Microsoft Azure, configuring virtual networks, subnets, NSGs, and authentication services.
- Performed pre-migration security hardening including system patching and service / port reduction.
- Designed cloud architecture across 20+ VMs using Azure VNet segmentation and NSGs to enforce least-privilege access control.

Accountabuddy | *React Native, Firebase, TypeScript*

January 2026 – May 2026

- Built a mobile application that improves task completion rates by introducing social accountability through shared task management.
- Engineered social features enabling users to add friends, share tasks, and track progress across 100+ active users.
- Utilized Firebase to manage backend functionality, enabling task updates, commenting, and user interactions.

Rubix Tutor | *Python, NumPy, Matplotlib*

August 2025 – January 2026

- Implemented an AI based Rubik's Cube solver using IDA* and IDDFS to search a large state space and generate optimal move sequences.
- Designed an admissible heuristic with a corner orientation pattern database, decreasing average solve depth.
- Modeled cube corner orientation and precomputed a pattern database with BFS to guide search.
- Developed a visualization interface with Matplotlib to display cube states and AI recommended moves.

TECHNICAL SKILLS

Languages: HTML, CSS, JavaScript, TypeScript, Python, C++, Bash, SQL

Frameworks / Libraries: Node.js, Express.js, React, React Native, NumPy, Matplotlib

Databases: Firebase, MySQL, MongoDB

Developer Tools: Microsoft Azure, Git, GitHub, Postman, MongoDB Compass, Vercel, Visual Studio Code, Linux